Supporting Information for

Strong Enhancement of Raman Scattering from a Bulk-Inactive Vibrational Mode in Few-layer $MoTe_2$

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S1. The Γ-point phonon modes of bulk $2H_b$-MX_2

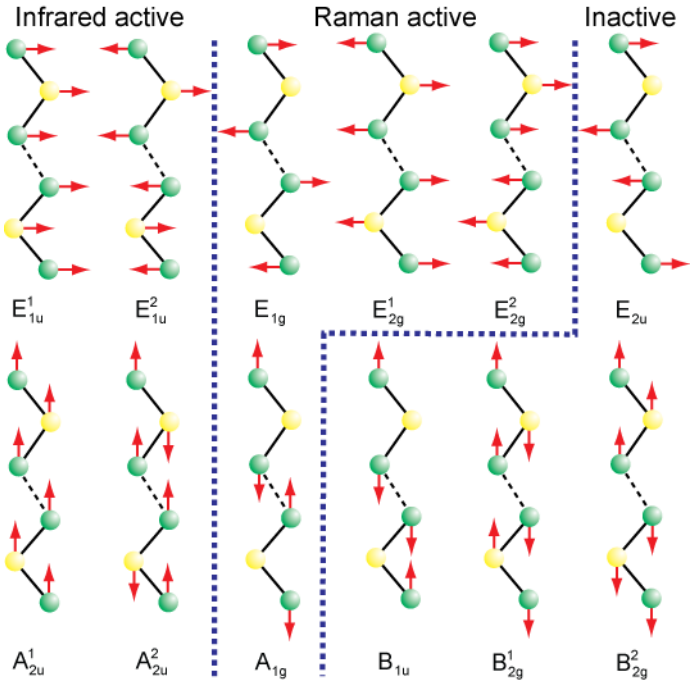


Figure S1. Phonon modes of bulk $2H_b$-MX_2 at the Γ point. Phonons on the top and bottom rows are the in-plane and out-of-plane vibrational modes, respectively. The black dashed lines indicate interlayer interactions. E^1_{1u}, E^2_{1u}, A^1_{2u}, and A^2_{2u} are infrared active modes, E_{1g}, E^1_{2g}, E^2_{2g}, and A_{1g} are Raman-active modes, and E_{2u}, B_{1u}, B^1_{2g}, and B^2_{2g} are optically inactive modes. (E^1_{1u}, E^2_{2g}), (E^2_{1u}, E^1_{2g}), (E_{1g}, E_{2u}), (A^1_{2u}, B^2_{2g}), (A^2_{2u}, B^1_{2g}), and (A_{1g}, B_{1u}) are conjugate pairs with slightly different frequencies under the interlayer interactions.

S2. Identification of the number of layers of $MoTe_2$ from the Raman intensity ratios

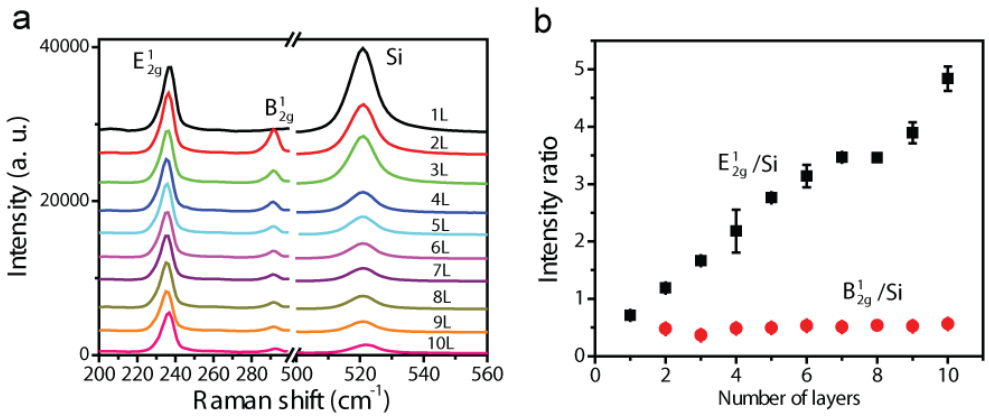


Figure S2. (a) Raman spectra of single- to 10-layer $MoTe_2$ supported on Si with top oxide layers with thicknesses of 90 nm. The intensities are normalized with the E^1_{2g} peak intensity in single-layer $MoTe_2$. (b) Intensity ratios of the E^1_{2g} (black squares) and B^1_{2g} (red circles) peaks of $MoTe_2$ to the Si peak.

Previous Raman spectroscopy studies of layered materials supported on a SiO_2/Si substrate have shown that the relative intensities of the peaks from the flakes to the Si peak increase with increasing the number of layers. Accordingly, the peak intensity ratios can be used to identify the thickness of the layered materials.[1] Here, we probe the intensity ratio of the E^1_{2g} peak of $MoTe_2$ to the Si peaks to confirm the number of layers determined by atomic force microscopy and optical contrast differences. Figure S2a shows Raman spectra of single- to 10-layer $MoTe_2$ on SiO_2/Si substrates. For each layer, the intensity of the Raman spectrum is normalized with the intensity of the E^1_{2g} peak of single-layer $MoTe_2$. We find the relative intensity of the Si to E^1_{2g} peak decreases with increasing thickness, as observed in MoS_2.[1] In Fig. S2b, we plot the intensity ratios of the E^1_{2g} and B^1_{2g} peaks to the Si peak as functions of the number of layers. The intensity ratio of the E^1_{2g} to Si peaks shows clear thickness-dependence. The relatively small intensity ratio at eight-layer thickness is due to the effects of small sample sizes, which lead to decrease in the E^1_{2g} peak intensity relative to the Si peak intensity. In contrast, the intensity ratio of the B^1_{2g} to Si peaks is nearly constant from bi- to 10-layers, since the intensity of the B^1_{2g} peak is enhanced with decreasing thickness, in contrast with the bulk Raman active modes, as discussed in the main text.

S3. Optical contrast differences between atomically thin $MoTe_2$ and SiO_2 surfaces

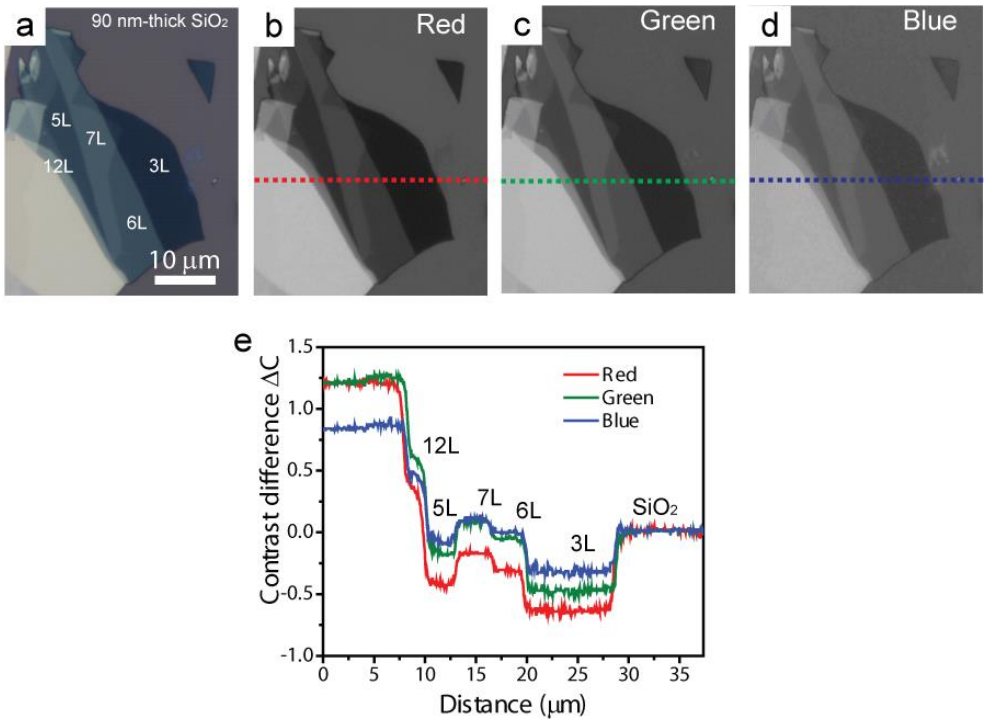


Figure S3. (a) Typical optical image of thin $MoTe_2$ films on 90 nm-thick SiO_2 and its gray scale images for the (b) red, (c) green, and (d) blue channels. The number of layers (NL with N = 3, 5, 6, 7, 12) is indicated. (e) Contrast profiles along lines shown in (b-d).

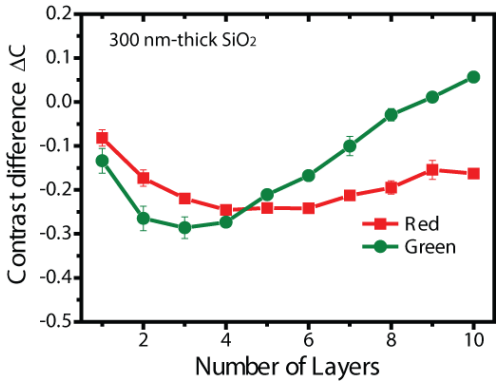


Figure S4. Optical contrast differences between $MoTe_2$ and 285 nm-thick SiO_2 surfaces. The green channel shows clearer thickness-dependence than the red channel.

The normalized optical contrast difference between $MoTe_2$ and SiO_2 surfaces can be defined as $\Delta C = (C_{MoTe2} - C_{SiO2})/C_{SiO2}$, where C_{MoTe2} and C_{SiO2} are the optical

contrasts of the $MoTe_2$ flakes and the SiO_2 substrates for each channel.[2] Figures S3a-d are bare optical and its gray-scale images for the red, green, and blue components. Figure S3e shows profiles along lines shown in Figs. S3b-d. The profiles show clearly the layer number differences for each color channel.

S4. Peak positions and intensities of the E^1_{2g} and A_{1g} modes and their thickness-dependences for MoS_2, $MoSe_2$, and $MoTe_2$

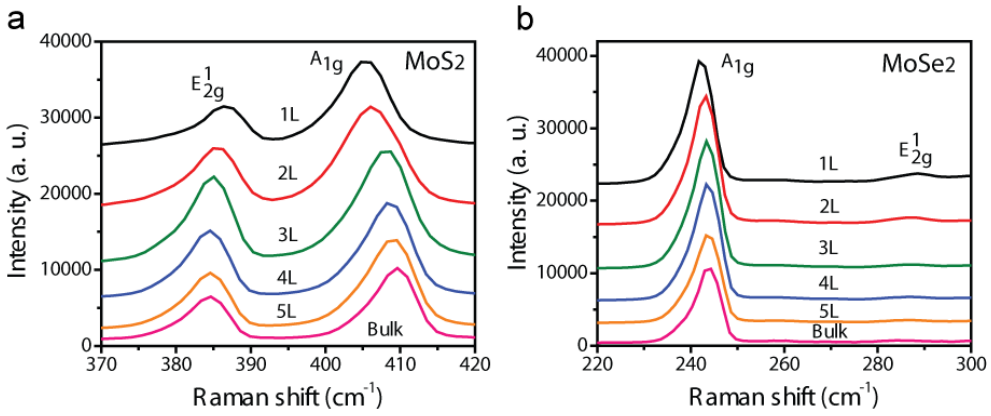


Figure S5. (a and b) Raman spectra of (a) MoS_2 and (b) $MoSe_2$ with various thicknesses. The peaks of the E^1_{2g} and A_{1g} modes are observed in both MoS_2 and $MoSe_2$.

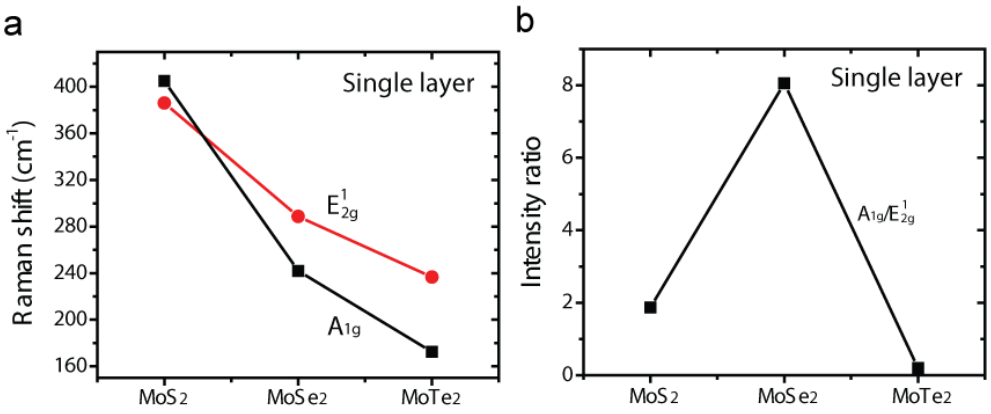


Figure S6. (a) Peak positions of the E^1_{2g} and A_{1g} modes of single-layer MoS_2, $MoSe_2$, and $MoTe_2$. The frequencies of the modes decrease with increasing molecular weight, as predicted by the DFT calculations.[3] (b) Intensity ratios of the A_{1g} to E^1_{2g} peaks of single-layer MoS_2, $MoSe_2$, and $MoTe_2$. The different intensity ratios among these compounds may result from the different polarizability of the chalcogen atoms.

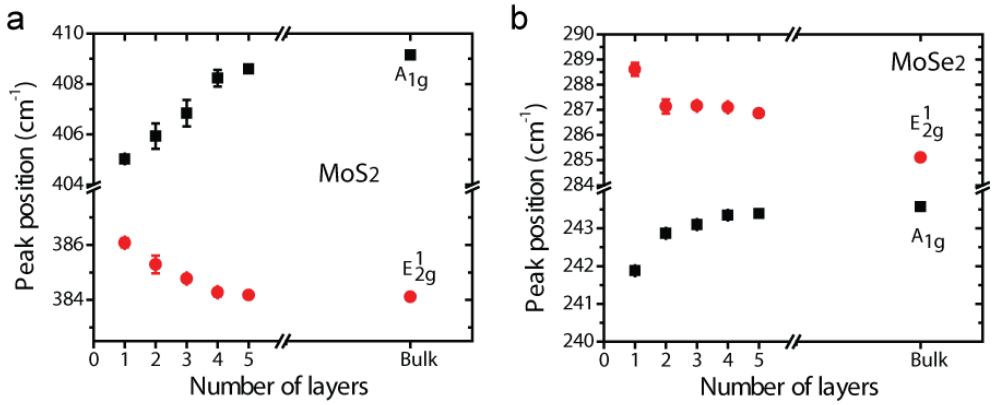


Figure S7. Peak positions of the E^1_{2g} and A_{1g} modes as functions of thickness for (a) MoS_2 and (b) $MoSe_2$. In MoS_2, the E^1_{2g} peak upshifts by ~ 2 cm^{-1}, while the A_{1g} peak downshifts by ~ 4 cm^{-1} with decreasing thickness from bulk to single-layer. In $MoSe_2$, the E^1_{2g} peak upshifts by ~ 3 cm^{-1}, while the A_{1g} peak downshifts by ~ 2 cm^{-1} with decreasing thickness from bulk to single-layer.

S5. The 633 nm-excited Raman spectrum of atomically thin $MoTe_2$

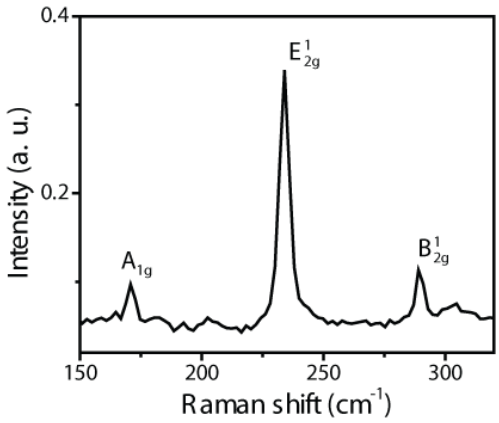


Figure S8. Raman spectrum of bilayer $MoTe_2$ on 285 nm-thick SiO_2. The excitation wavelength is a 633 nm. The B^1_{2g} peak is clearly observed at ~ 291 cm^{-1}.

S6. The calculated phonon frequencies of $MoTe_2$

Table S1. Density functional theory-calculated frequencies (in cm^{-1}) of the in-plane phonon modes in single- to four-layer and bulk crystals of α-$MoTe_2$. "R", "IR", and "Ina" indicate Raman and infrared activity of the modes.

	In-plane					
Bulk	E^1_{1u} (IR) 0	E^2_{2g} (R) 29.4	E_{2u} (Ina) 119.3	E_{1g} (R) 121.6	E^2_{1u} (IR) 239.8	E^1_{2g} (R) 239.9
1L	E' (R+IR) 0		E'' (R) 120.2		E' (R+IR) 242.3	
2L	E_u (IR) 0.1	E_g (R) 21.0	E_u (IR) 119.8	E_g (R) 121.0	E_u (IR) 241.0	E_g (R) 241.0
3L	E' (R+IR) 0.1	E' (R+IR) 25.6 E'' (R) 15.1	E''(R) 121.4 E' (R+IR) 120.5 E'' (R) 119.7		E' (R+IR) 241.0 E'' (R) 241.0 E' (R+IR) 239.9	
4L	E_u (IR) 0.3	E_g (R) 26.9 E_u (IR) 20.2 E_g (R) 10.0	E_g (R) 121.3 E_u (IR) 120.6 E_g (R) 119.8 E_u (IR) 119.4		E_u (IR) 240.9 E_g (R) 240.9 E_g (R) 239.9 E_u (IR) 239.9	

Table S2. Density functional theory-calculated frequencies (in cm^{-1}) of the out-of-plane phonon modes in single- to four-layer and bulk crystals of α-$MoTe_2$.

	Out-of-plane					
Bulk	A^1_{2u} (IR) 0	B^2_{2g} (Ina) 39.3	B_{1u} (Ina) 176.3	A_{1g} (R) 179.9	A^2_{2u} (IR) 290.4	B^1_{2g} (Ina) 296.9
1L	A_2" (IR) 0		A_1' (R) 178.1		A_2" (IR) 300.5	
2L	A_{2u} (IR) 0.7	A_{1g} (R) 31.4	A_{2u} (IR) 177.9	A_{1g} (R) 180.2	A_{2u} (IR) 297.7	A_{1g} (R) 298.7
3L	A_2" (IR) 0.7	A_2" (IR) 34.1 A_1' (R) 20.7	A_1' (R) 179.4 A_2" (IR) 178.3 A_1' (R) 177.0		A_1' (R) 298.5 A_2" (IR) 298.2 A_2" (IR) 295.9	
4L	A_{2u} (IR) 0.5	A_{1g} (R) 36.5 A_{2u} (IR) 28.4 A_{1g} (R) 16.5	A_{1g} (R) 179.6 A_{2u} (IR) 178.8 A_{1g} (R) 177.8 A_{2u} (IR) 176.7		A_{1g} (R) 298.5 A_{2u} (IR) 298.3 A_{1g} (R) 296.6 A_{2u} (IR) 295.4	

S7. Raman spectra of suspended- and supported-$MoTe_2$

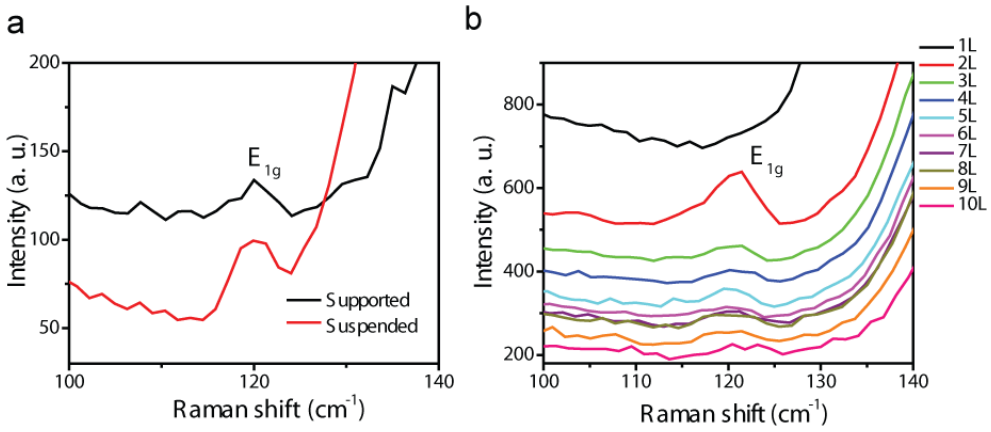


Figure S9. (a) Raman spectra of trilayer $MoTe_2$ supported on SiO_2 (black line) and suspended over a hole with a diameter of 2 μm (red line). The Raman spectra show the E_{1g} peaks at ~ 120 cm^{-1}, though the E_{1g} mode is forbidden in the backscattering configuration. The activation of the E_{1g} peaks is due to translation symmetry breaking rather than substrate effects. The enhancement of the peak intensity in the suspended sample is likely due to the absence of the coupling to the substrate.[4] (b) Raman spectra of single-layer (1L) to 10-layer (10L) $MoTe_2$ supported on SiO_2. The E_{1g} peak is absent in single-layer $MoTe_2$, but present in few-layer $MoTe_2$. The peak intensity decreases with increasing thickness. These observations are consistent with the previous reports on WSe_2.[5]

References

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